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Amrita School of Computing, Amritapuri Campus
Department of Computer Science and Engineering

B.Tech CSE — Initial Project Idea Submission

Project Team

Roll Number	Name
AM.SC.U4CSE23147	Rishi Priyan
AM.SC.U4CSE23159	Sujith Matam
AM.SC.U4CSE23172	Pranav P Nair

Guide Details

Name	Department	Contact Number
Ms. Gopika Muraleedas	School of Computing	+91 81292 62678
Dr. Divya Udayan	School of Computing	+91 95507 97705

1. Project Title

Project Title:

An Agentic AI-Powered Orthopedic Postoperative Chatbot with Multi-modal Multi-Agent Collaboration

2. Project Abstract

2.1 Motivation

Motivation:

Orthopedic surgeries — including joint replacements, sports medicine procedures, and fracture repairs — affect millions of patients annually worldwide. Effective postoperative care is critical to surgical outcomes; however, traditional doctor-led follow-up systems are severely constrained by limited physician availability, high consultation costs, and the inability to provide continuous 24/7 support. Patients recovering at home frequently experience anxiety, uncertainty about rehabilitation exercises, and delayed responses to complications, all of which adversely impact recovery.

A randomized controlled trial published in *npj Digital Medicine* (Li et al., 2026) demonstrated that a GPT-4–powered AI agent on WeChat significantly improved patient satisfaction, knowledge scores, and early functional recovery over conventional doctor-led follow-up. However, that system depended on a monolithic single-agent LLM tethered to a proprietary cloud API — making it unsuitable for privacy-sensitive clinical environments where patient data must remain on-premise.

Concurrent advances in agentic AI further motivate this work. Meissa (Chen et al., 2026) demonstrated that lightweight 4B-parameter multi-modal medical agents, trained via agentic behavior distillation from frontier models, can match or exceed proprietary frontier agents across 13 medical benchmarks while operating fully offline with 22× lower latency. The Agentic AI System for Stress Monitoring (Hosseini & Seilani, 2026) showed that multi-agent CrewAI frameworks with physiological sensor integration can autonomously monitor patient health with adaptive personalized thresholding. Together, these works establish a compelling foundation for building a **lightweight, offline-capable, multi-modal agentic AI system** for orthopedic postoperative follow-up.

2.2 Problem Statement

Problem Statement:

Existing postoperative follow-up AI systems are either: (a) cloud-dependent single-agent LLMs that cannot be deployed on-premise for patient data privacy reasons, (b) reactive single-turn chatbots lacking contextual awareness, personalized rehabilitation guidance, and proactive monitoring, or (c) narrow agent frameworks not validated in orthopedic postoperative care. Current systems cannot autonomously plan recovery timelines, interpret physiological sensor data, coordinate multi-agent clinical reasoning, or adapt to individual patient recovery trajectories — while remaining deployable offline in resource-constrained clinical environments.

The proposed project addresses these gaps by designing a **lightweight, offline-deployable, multi-modal agentic AI system** that: autonomously handles postoperative patient queries through an LLM-based chatbot interface; performs continuous real-time physiological monitoring via wearable sensors; applies agentic behavior distillation (inspired by Meissa) to train a compact model capable of tool-augmented reasoning, multi-agent collaboration, and adaptive clinical simulation — without requiring cloud API access during inference.

3. Objectives

Objectives:

1. **Design** a multi-agent agentic AI framework comprising specialized agents for patient intake, rehabilitation guidance, physiological signal monitoring, anomaly detection, and automated clinical report generation, tailored for orthopedic postoperative care.

2. **Develop** a lightweight, offline-deployable conversational chatbot model using agentic behavior distillation techniques (as in Meissa) that matches frontier LLM performance on postoperative medical QA while operating without cloud API dependency.

3. **Implement** a real-time physiological stress and recovery monitoring module using wearable PPG sensor data with adaptive, patient-specific thresholding for early detection of abnormal recovery patterns.

4. **Evaluate** the proposed system against single-LLM baselines, cloud-based systems, and conventional follow-up methods using metrics including response accuracy, patient satisfaction, recovery outcome indicators, clinical report quality, and end-to-end inference latency.

5. **Analyze** the research gaps identified from the five base papers — including cloud dependency, absence of orthopedic-specific personalization, lack of physiological sensor integration, and absence of offline agentic benchmarking for postoperative care — and validate that the proposed system addresses these gaps.

4. Dataset / Data Source

Dataset / Data Source:

1. **Stress-Predict Dataset** (Publicly available) — Physiological data from 35 participants collected using Empatica E4 PPG wrist-worn sensors, annotated with stress/non-stress labels using Perceived Stress Scale (PSS) and State-Trait Anxiety Inventory (STAI) questionnaires. Used for training and evaluating the physiological monitoring module.
   URL: `https://github.com/italha-d/Stress-Predict-Dataset`

2. **Orthopedic Patient Q&A Interaction Logs** — Patient inquiry data from the RCT conducted by Li et al. (2026), categorized into symptom consultation, rehabilitation guidance, medication queries, and postoperative recovery, used to train and evaluate the chatbot dialogue component.

3. **Medical Agentic Trajectory Dataset (Meissa)** — The ˜40K curated frontier multimodal agent trajectories (across radiology, pathology, clinical reasoning benchmarks) released by Chen et al. (2026) at `https://github.com/Schuture/Meissa`, used for behavior distillation of the lightweight offline model.

4. **Synthetic Orthopedic Conversation Dataset** — To be generated via LLM-assisted data augmentation for orthopedic-specific postoperative dialogues, following practices from MedAgentBoard (Zhu et al., 2025).

5. Expected Deliverables

Expected Deliverables:

1. A **lightweight offline-deployable multi-modal agentic AI system prototype** for orthopedic postoperative follow-up, trained via behavior distillation and capable of tool-augmented reasoning, multi-agent collaboration, and clinical simulation without cloud API access.

2. A **conversational chatbot interface** supporting real-time, personalized, context-aware patient interaction with postoperative guidance through a multi-turn dialogue system.

3. A **physiological signal monitoring and anomaly detection module** integrating wearable PPG sensor data for continuous stress and recovery monitoring with patient-adaptive thresholding and automated LLM-driven clinical report generation.

6. Preliminary Literature Survey

No.	Paper Title	Year	Key Contribution
1	A Randomized Controlled Trial of a WeChat-Based Artificial Intelligence Agent for Postoperative Care in Orthopedic Patients	2026	Demonstrated via an RCT that a GPT-4–powered AI agent on WeChat significantly improved patient satisfaction, knowledge scores, and early functional recovery over doctor-led follow-up in orthopedic postoperative care.
2	Meissa: Multi-modal Medical Agentic Intelligence	2026	Introduced a 4B-parameter lightweight multi-modal medical agent trained via three-tier agentic behavior distillation from frontier models across ~40K trajectories; matches or exceeds frontier models on 10 of 16 settings across 13 medical benchmarks with 22× lower latency and full offline capability.
3	Agentic AI System for Stress Monitoring	2026	Proposed a multi-agent agentic AI framework (CrewAI) with PPG-based physiological stress classification, adaptive personalized thresholding, and LLM-driven clinical report generation, enabling autonomous on-device health monitoring with minimal human oversight.
4	Large Action Models: From Inception to Implementation	2024	Introduced Large Action Models (LAMs) — AI systems that execute real-world actions beyond text generation — and presented a complete development pipeline covering data collection, training, agent integration, grounding, and evaluation using a Windows GUI agent (UFO) as a case study.
5	MedAgentBoard: Benchmarking Multi-Agent Collaboration with Conventional Methods for Diverse Medical Tasks	2025	Introduced a comprehensive medical AI benchmark evaluating multi-agent collaboration, single-LLM, and conventional methods across four task categories (medical QA, EHR predictive modeling, lay summary, and clinical workflow automation), revealing nuanced, context-dependent advantages of multi-agent collaboration.

6.1 Comparison of Related Work

No.	Paper / Ref.	Year	Methodology	Dataset Used	Limitations / Research Gap
1	Li et al. [1]	2026	Single-agent GPT-4 LLM chatbot on WeChat; RCT evaluation (AI vs. doctor-led)	Orthopedic patient RCT data (single-center, China); GAD-7, PCS, MCS, IKDC, FJS questionnaires	Cloud-dependent single-agent; restricted platform/language; no physiological monitoring; no multi-agent collaboration; long-term convergence
2	Chen et al. [2]	2026	Agentic behavior distillation via 3-tier stratified trajectory supervision; prospective-retrospective training	~40K curated trajectories (from Gemini-3-flash); 13 medical benchmarks (radiology, pathology, clinical QA)	General medical imaging focus; not tested for postoperative care/rehabilitation; no physiological sensor integration
3	Hosseini & Seilani [3]	2026	Multi-agent CrewAI framework; XGBoost/RF PPG classifier; adaptive dynamic thresholding; LLM clinical reports	Stress-Predict dataset (35 subjects with PPG E4 sensor, PSS/STAI labels)	Confined to stress monitoring; no postoperative care dialogue; no conversational patient interface; small dataset
4	Wang et al. [4]	2024	LAM pipeline: GUI action data collection, LLM action fine-tuning, agent system integration (UFO GUI case study)	Windows GUI task datasets; synthetic/real user request logs for GUI interactions	Focused on GUI automation; clinical care not addressed; safety and privacy challenges open; no clinical validation
5	Zhu et al. [5]	2025	Benchmark (MedAgentBoard) comparing multi-agent systems vs. single-LLMs and conventional ML models	MedQA, PubMedQA, PathVQA, VQA-RAD, MIMIC-III EHR, eICU, and clinical workflow datasets	No orthopedic or postoperative tasks; no patient-facing conversational dialogue; no offline agent benchmarking

Table 1: Comparison of related work across key aspects.

6.2 Novelty / Research Gap Statement

Novelty / Research Gap:

The five papers that were reviewed implement the state of art in AI-assisted postoperative care, physiological monitoring, multiagent medical benchmarking, action-oriented AI, lightweight medical agentic intelligence, although none of them looks into the specific combination that is necessary for the given project.

The benefit of an LLM orthopedic chatbot was shown by Li et al. (2026) by using a single-agent architecture dependent on cloud, but it had no physiological awareness. It's advantages in comparison to doctor-based care diminished at approximately six months. Chen et al. (2026) used behavior distillation to introduce an effective offline-capable lightweight medical agent, but they only tested it on conventional medical imaging and quality assurance standards; they excluded physiological sensors, postoperative care, or rehabilitation talks. Multi agentic psychological monitoring was implemented by Hosseini & Seilani (2026). It had agentic autonomy, but it did not give any communicational patient interface and did not extend their structure/framework to postoperative orthopedic settings. Wang et al. (2024) set the stage for action-based LAMs implemented them only to GUI environments which were digital. He did not give any context clinically nor regarding patient data. Zhu et al. (2025) strictly evaluated the performance of multi agent medical collaboration but failed to review postoperative, orthopedic-specific, or wearable integrated environments

The proposed project fills all these gaps by incorporating: (1) **Agentic behavior distillation** which was inspired by Meissa to train an offline model which is deployable for postoperative dialogue; (2) **Multi–agent orchestration** with specialized agents that handle queries, rehabilitation, and anomaly detection.; (3) **Real-time PPG-based physiological monitoring** features adaptive patient-specific thresholding; and finally, (4) **Orthopedic specific postoperative evaluation** that belongs in the CT framework of Li et al. (2026) - all inside a offline privacy secured, capable system. This integration/incorporation is unaddressed in any prior work.

7. Potential Publication Venues

Possible Conferences

NeurIPS, ICML, ICLR, AAAI, IJCAI, KDD, ICDM, SDM, AISTATS, UAI, CVPR, ICCV, ECCV, WACV, BMVC, ACL, EMNLP, NAACL, COLING, ECAI, ICPR, ICIP, ICASSP, COMSNETS, CODS-COMAD, INDICON, ICACCS, ICCCI, IEEE ANTS, COMPUTE, ICACC, TENCON

Possible Journals

IEEE Access, Expert Systems with Applications, Knowledge-Based Systems, Applied Soft Computing, Engineering Applications of Artificial Intelligence, Information Sciences, Pattern Recognition, Neurocomputing, Artificial Intelligence Review, Machine Learning, Neural Computing and Applications, Journal of King Saud University – Computer and Information Sciences, Multimedia Tools and Applications, Computers & Security, Computer Networks, Future Generation Computer Systems, Computer Methods and Programs in Biomedicine, Artificial Intelligence in Medicine, Data & Knowledge Engineering, ACM Transactions on Intelligent Systems and Technology (TIST), ACM Computing Surveys, IEEE Transactions on Artificial Intelligence, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Knowledge and Data Engineering (TKDE), IEEE Transactions on Dependable and Secure Computing (TDSC), Scientific Reports, PLOS ONE, Sensors, Electronics, Heliyon

Selected Venues & Justification:

Conferences:

- **AAAI** — Premier AI venue directly covering multi-agent systems, agentic AI, behavior distillation, and healthcare AI.
- **CODS-COMAD** — Leading Indian data science and AI conference; appropriate for applied LLM and agentic healthcare systems.
- **ICACC** — National-level conference in India on advances in computing; suitable for applied healthcare AI prototypes and system demonstrations.

Journals:

- **npj Digital Medicine** (Nature Portfolio) — Published base paper (Li et al., 2026); directly aligned with AI-assisted postoperative care and digital health research.
- **Artificial Intelligence in Medicine** (Elsevier) — High-impact journal covering AI, LLMs,

agentic systems, and clinical AI applications.

- **IEEE Access** — Broad open-access venue for applied AI and healthcare systems; indexed in major databases.

8. References

[1] J. Li, Y. Zhang, Z. Zhang, Y. Zhou, Y. Gao, X. Li, and S. Fan, "A randomized controlled trial of a WeChat-based artificial intelligence agent for postoperative care in orthopedic patients," *npj Digital Medicine*, vol. 9, Art. no. 105, 2026, doi: 10.1038/s41746-025-02269-8.

[2] H. Chen, Y. Hou, M. Zhong, et al., "Meissa: Multi-modal medical agentic intelligence," *arXiv preprint arXiv:2603.09018*, 2026. Available: https://github.com/Schuture/Meissa.

[3] M. Hosseini and H. Seilani, "Agentic AI system for stress monitoring," *International Journal of Computational Intelligence Systems*, vol. 19, Art. no. 95, 2026, doi: 10.1007/s44196-026-01215-0.

[4] J. Wang, Z. Xu, X. Zhao, et al., "Large action models: From inception to implementation," *arXiv preprint arXiv:2412.10047*, 2024.

[5] Y. Zhu, Z. He, H. Hu, X. Zheng, X. Zhang, Z. Wang, J. Gao, L. Ma, and L. Yu, "MedAgentBoard: Benchmarking multi-agent collaboration with conventional methods for diverse medical tasks," *arXiv preprint*, 2025.

8. Faculty Remarks

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